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title: "Machine Learning Assignment 4"
output: html_document
date: "2026-10-27"

```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

#Imported the data set.
```{r}
Pharmaceuticals <- read.csv("~/Downloads/Pharmaceuticals.csv")
head(Pharmaceuticals)
```

#I used these codes to get the size, column, names, structure and summary of a dataset.
```{r}
dim(Pharmaceuticals)
names(Pharmaceuticals)
str(Pharmaceuticals)
summary(Pharmaceuticals)
```

#I am checking to see if there is missing values present.
```{r}
colSums(is.na(Pharmaceuticals))
```

#I am selecting the 9 main variables needed to do clustering on different scales. The attributes have different values so we wanted to put them on the same range.
```{r}
financial_data <- Pharmaceuticals[, c(
  "Market_Cap",
  "Beta",
  "PE_Ratio",
  "ROE",
  "ROA",
  "Asset_Turnover",
  "Leverage",
  "Rev_Growth",
  "Net_Profit_Margin"
)]
head(financial_data)
```

#I standardized the nine financial variables because they are measured on different scales. Standardization gives each variable a mean of zero and a standard deviation of one, preventing variables with larger numerical scales from dominating the clustering. I gave each standardized variable equal weight.
```{r}
financial_scaled <- scale(financial_data)
head(financial_scaled)
```

#I am creating an elbow plot to help decide how many clusters to use. The plot shows the sum of squares which shows how close the companies are grouped with their range.
```{r}
set.seed(123)
```

```
wss <- numeric(6)

for (k in 1:6) {
  model <- kmeans(
    financial_scaled,
    centers = k,
    nstart = 25
  )

  wss[k] <- model$tot.withinss
}

plot(
  1:6,
  wss,
  type = "b",
  xlab = "Number of Clusters",
  ylab = "Within-Cluster Sum of Squares",
  main = "Elbow Plot"
)
```

#I am using k-means to group companies with similar financial characteristics.
```{r}
set.seed(123)

pharma_clusters <- kmeans(
  financial_scaled,
  centers = 3,
  nstart = 25
)

pharma_clusters
```

I am adding each company's cluster number to the original dataset.
```{r}
Pharmaceuticals$Cluster <- pharma_clusters$cluster
Pharmaceuticals[, c("Name", "Symbol", "Cluster")]
```

#I am creating the clusters and counting the companies for each one.
```{r}
set.seed(123)
pharma_clusters <- kmeans(
  financial_scaled,
  centers = 3,
  nstart = 25
)
Pharmaceuticals$Cluster <- pharma_clusters$cluster
table(Pharmaceuticals$Cluster)
```

#I am finding the average financial values for each cluster and to understand what makes them different.Looking at these avaerages helps to see if a cluster has larger companies, more profitable companies, or companies with higher growth or leverage.
```{r}
cluster_averages <- aggregate(
  financial_data,
  by = list(Cluster = Pharmaceuticals$Cluster),
  FUN = mean
)
round(cluster_averages, 2)
```

Cluster Names and Reasoning

Cluster 1: High P/E, Low-Profitability Companies

This cluster contains four companies. It has the highest average P/E ratio and the lowest average ROE, ROA, and net profit margin among the three clusters.

Cluster 2: Large, Profitable Companies

This cluster contains eleven companies. It has the highest average market capitalization, ROE, ROA, and net profit margin, along with below-average beta and leverage.

Cluster 3: Smaller, High-Growth, Highly Leveraged Companies

This cluster contains six companies. It has the lowest average market capitalization and the highest average revenue growth and leverage. Its profitability is below the overall average.

#I am checking to see which companies go for each cluster.

```{r}
Pharmaceuticals[, c("Name", "Symbol", "Cluster")]
```

#I am checking whether brokerage recommendations differ across clusters.

```{r}
table(
  Pharmaceuticals$Cluster,
  Pharmaceuticals$Median_Recommendation
)
```

#Brokerage recommendations differ somewhat across the clusters. In Cluster 1, Hold is the most common recommendation, covering 2 of 4 companies (50%). This cluster also contains the only company with a Strong Buy recommendation. In Cluster 2, Hold is also most common, covering 6 of 11 companies (54.5%). In Cluster 3, Moderate Buy is most common, covering 3 of 6 companies (50%). Therefore, Clusters 1 and 2 are mainly associated with Hold recommendations, while Cluster 3 has a larger proportion of Moderate Buy recommendations.

```{r}
table(
  Pharmaceuticals$Cluster,
  Pharmaceuticals$Location
)
```

#I am checking whether stock exchanges differ across clusters. Headquarters locations show some differences across the clusters. Cluster 1 is geographically spread out, with one company each in Canada, Germany, the UK, and the US. Clusters 2 and 3 are mainly US-based: 8 of 11 companies (72.7%) in Cluster 2 and 4 of 6 companies (66.7%) in Cluster 3 are headquartered in the US. Cluster 1 therefore has a more evenly distributed geographic composition, while the other two clusters are concentrated in the US.

```{r}
table(
  Pharmaceuticals$Cluster,
  Pharmaceuticals$Exchange
)
```

#All four companies in Cluster 1 and all eleven companies in Cluster 2 are listed on the NYSE. Cluster 3 has four companies listed on the NYSE, one on AMEX, and one on NASDAQ. The

NYSE is the most common exchange in every cluster, but Cluster 3 contains all the AMEX and NASDAQ companies in this dataset. Exchange therefore provides some distinction for Cluster 3, but it does not distinguish Clusters 1 and 2.